

Long Multiplication Journey

Grade 4 Math Lesson

What is Long Multiplication?

Long multiplication is a special method that helps us multiply bigger numbers easily! When we need to multiply two 2-digit numbers, it can seem tricky. But with long multiplication, we break it down into simple steps: multiplying one digit at a time and then adding them together.

1. Understanding Place Values

Before we start, remember how numbers are built:

- **Ones Place:** Tells us how many single items (e.g., in 34, the digit 4 is in the ones place).
- **Tens Place:** Tells us how many groups of 10 (e.g., in 34, the 3 means 30).

$$34 = 30 + 4$$

2. The Three Simple Steps

Step 1: Multiply by the Ones Digit

Multiply the ones digit of the second number by each digit of the first number, starting from the right. If the product is greater than 9, carry over the tens digit.

Example: 12×13

Multiply 12 by 3 (ones digit of 13).

$12 \times 3 = 36$ (Our first partial product).

Step 2: Multiply by the Tens Digit

Multiply the tens digit of the second number by the first number. **CRITICAL: Always add a zero as a placeholder** because you are multiplying by tens!

Example: 12×13

Multiply 12 by 1 (tens digit of 13). Add the zero.

$12 \times 10 = 120$ (Our second partial product).

Step 3: Add Partial Products

Line up your partial products by place value (ones under ones, tens under tens) and add them up.

$$36 + 120 = 156$$

So, $12 \times 13 = 156$.

3. Real-World Example

Imagine one packet of chips costs 34 and you want to buy 26 packets for a party.

- **Step 1 (34×6):** 204
- **Step 2 (34×20):** 680
- **Step 3 (Add):** $204 + 680 = 884$

4. Common Mistakes to Avoid

- **Forgetting Zero:** Always add the zero placeholder in Step 2!
- **Misalignment:** Line up numbers neatly by their place value.
- **Skipping Carry Overs:** Remember to add the digit you carried over.

5. Practice Problems

Try these in your notebook before checking the logic:

1. 23×15 (Hint: $115 + 230$)
2. 45×12 (Hint: $90 + 450$)
3. 52×24 (Hint: $208 + 1040$)